

BREAK INTO AEROSPACE

From Laboratory Technician to Aerospace

Career Transition Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: Laboratory technicians are among the most directly qualified candidates for some of the most safety-critical technical roles in aerospace — and among the most consistently overlooked by aerospace recruiters who have not thought carefully about what those roles actually require. The laboratory technician who has spent five years running tensile tests, measuring hardness, preparing metallographic sections, and writing test reports has developed the specific competencies — precision measurement, rigorous documentation, specimen preparation, data interpretation, calibration management, and quality system adherence — that aerospace materials testing, NDT, and quality engineering roles depend on. The aerospace sector adds specific material knowledge, specific regulatory standards, and specific safety consequences to work that is methodologically familiar. The translation is specific, learnable, and bounded.

This guide is written for people who have built their careers in laboratory environments through vocational routes — HNC or HND qualifications, BTEC programmes, RSCC or City & Guilds technical certificates, laboratory assistant or technician apprenticeships, science technician roles, or NVQ-based laboratory pathways — and who want to move into the aerospace sector. It is explicitly not a guide for materials scientists or chemists with degrees: those guides exist separately and address different preparation challenges. This guide is for the person who runs the test rather than designs the test method, who measures and documents rather than derives and analyses, and who understands the quality system from the inside because they work within it every day.

The honest starting point about scope: the laboratory technician's transition to aerospace is not as narrow as it might first appear, but it is specifically targeted. Unlike the electrical technician, whose practical skills apply across almost the entire aerospace maintenance landscape, the laboratory technician's most natural aerospace home is in one of four concentrated domains: materials and mechanical testing (tensile, compressive, fatigue, hardness, impact, and fracture toughness testing of metals and composites to aerospace standards), NDT inspection (the non-destructive testing of aerospace components using ultrasonic, eddy current, dye penetrant, magnetic particle, and radiographic methods), aerospace quality and inspection (first article inspection, in-process inspection, receiving inspection, and metrology in aerospace manufacturing), and chemical analysis and fuel quality (analytical chemistry applied to aviation fuel testing, propellant analysis, or aerospace coating and fluid characterisation). Each of these is a substantive career with specific qualification pathways, specific regulatory frameworks, and specific skill development trajectories that the laboratory technician is better placed to enter than most other transitioning backgrounds.

The aerospace roles most accessible from a laboratory technician background: NDT technician (Level I and II certification under ASNT SNT-TC-1A or EN ISO 9712, the most directly accessible and largest aerospace laboratory role category); aerospace materials testing technician (tensile, hardness, fatigue, impact, and microscopy testing to ASTM and AMS standards at NADCAP-accredited test laboratories); aerospace quality and metrology technician (dimensional inspection, CMM operation, first article inspection, and calibration management in aircraft and spacecraft manufacturing); aviation fuel quality technician (analytical chemistry testing of Jet A/A-1 and SAF to ASTM D1655 and DEF STAN 91-091); aerospace environmental test technician (conducting DO-160 environmental tests — temperature, vibration, humidity, and EMI — on avionics equipment at certification test facilities); composite process technician (performing and quality-controlling the autoclave cure, lay-up, and inspection of composite structures under NADCAP accreditation); and propellant and energetic materials analysis technician (for clearance-eligible candidates targeting defence propulsion chemistry laboratories).

The employer landscape is broad: NADCAP-accredited materials test laboratories (Element Materials Technology, Intertek, Exova/Applus+, TIMET test laboratories, Alcoa test facilities), aerospace OEM in-house test laboratories (Rolls-Royce materials test, GE Aerospace materials characterisation, Airbus materials), MRO NDT inspection departments (Lufthansa Technik NDT, Delta TechOps NDT, ST Engineering inspection), aviation fuel quality organisations (Air BP, Shell Aviation, Energy Institute JIG-accredited fuel testing laboratories), DO-160 environmental test houses (NTS Group, Element, National Technical Systems, Curtiss-Wright test facilities), aerospace composite process laboratories (GKN Aerospace composite testing, Safran composite process control), and spacecraft materials and test facilities (ESA ESTEC materials laboratory, NASA materials test facilities at Johnson and Goddard, satellite prime contractor test labs).

PART I: WHAT AEROSPACE EMPLOYERS ACTUALLY SEE IN A LABORATORY TECHNICIAN

The Genuine Advantages — What Transfers Directly

The laboratory technician's core advantage — and the one that is most consistently underestimated by both parties in the hiring process — is the combination of measurement rigour, documentation discipline, and quality system fluency that comes from working in an accredited laboratory environment. An ISO 17025-accredited testing laboratory operates under quality management requirements that in some respects exceed those of a typical industrial quality department: every test method must be documented, every piece of equipment must be calibrated and within its calibration interval, every test result must be traceable to SI units through an unbroken calibration chain, every uncertainty in measurement must be estimated and reported, and every anomalous result must be investigated and documented. The laboratory technician who has worked in this environment has developed measurement and documentation habits that aerospace quality systems specifically require.

Specimen preparation competence — the ability to prepare a material sample for examination without introducing artefacts that compromise the measurement — is one of the most directly transferable laboratory skills. The metallographer who has cut, mounted, ground, polished, and etched a metallographic section to reveal the microstructure without smearing, over-etching, or introducing false features is doing exactly the same quality of work that an aerospace failure investigation requires. The tensile specimen preparer who has machined a round specimen to BS EN ISO 6892-1 dimensional tolerances, verified the gauge length and cross-sectional area, and aligned the specimen in the test machine without introducing bending load is doing exactly the preparation quality that an aerospace A-basis or B-basis allowable determination requires. The NDT trainee who has prepared a dye penetrant test panel without contaminating the surface with fingerprints is demonstrating the contamination-awareness that aerospace NDT requires.

Test result interpretation at the level of “is this result consistent with what we expect for this material, or is something wrong?” — the quality gate judgement that every experienced laboratory technician develops — is directly applicable in aerospace. The hardness measurement that falls 10 HRC below the specification minimum on a titanium forging is immediately flagged by an experienced technician who knows what Ti-6Al-4V should measure after the specified heat treatment. The tensile specimen that fractures in the threaded grip rather than the gauge length is correctly flagged as an invalid test even before the load-extension curve is analysed. This first-line quality judgement — the ability to recognise when a result is anomalous and why — is exactly what aerospace test laboratories need from their technicians.

The Specific Gaps — What Does Not Transfer Without Work

The aerospace material specification framework. The laboratory technician who has tested materials to ASTM standards and issued test reports in a commercial testing environment has the testing competence. What they typically lack is the aerospace material specification framework that determines which ASTM method is required for a specific aerospace material, what the acceptance criteria are, and why those specific criteria were set. AMS 2430 specifies the requirements for shot peening of metallic components; AMS 2770 specifies the heat treatment of aluminium alloys; AMS 4928 specifies Ti-6Al-4V bar. Each AMS specification calls out specific test methods, specific acceptance criteria, and specific sampling requirements that differ from generic commercial testing. The laboratory technician must understand not just how to run the test but what specification governs the material they are testing and why the acceptance criterion is what it is.

The NADCAP accreditation framework. NADCAP (National Aerospace and Defense Contractors Accreditation Program) is the aerospace industry accreditation programme for special processes including materials testing laboratories (NADCAP NLT — Non-Destructive Testing and Laboratory Testing). ISO 17025 accreditation — which many commercial testing laboratories hold — is a necessary but not sufficient credential for aerospace materials testing: NADCAP NLT accreditation adds aerospace-specific process control requirements (specific calibration standards, specific reference standards, specific documentation requirements for aerospace test reports) that go beyond ISO 17025. The laboratory technician who has worked in an ISO 17025 laboratory has a strong quality management foundation; the additional aerospace-specific requirements of NADCAP NLT need to be specifically studied.

The aerospace consequence of measurement error. In a commercial testing laboratory, a test result that is

later found to be incorrect due to specimen preparation error, equipment fault, or operator error results in a retested specimen and a corrected report. The customer is inconvenienced. In an aerospace materials testing context, an incorrect A-basis tensile strength that has been incorporated into an aircraft structural analysis may result in a structural design that does not have the required safety margin — a consequence that may not become apparent until a structural failure occurs. The laboratory technician entering aerospace must specifically develop the understanding that their test results are not data points in a report but inputs to engineering decisions with safety consequences. This does not require the technician to become a structural engineer — but it does require them to understand why the NADCAP auditor cares so much about the calibration of the extensometer and the dimensional accuracy of the specimen.

The NDT certification and regulatory framework. NDT in aerospace is a specific regulatory domain — ASNT SNT-TC-1A in the US (employer-based certification) and EN ISO 9712 in Europe (third-party certification) — with specific training hour requirements, specific examination content, and specific recertification intervals. The laboratory technician who has used NDT methods in a quality control context — dye penetrant to inspect welds, ultrasonic to check a casting — has valuable practical exposure but not the formally certificated qualification that aerospace NDT roles require. Understanding the specific path from practical exposure to formal certification is the regulatory knowledge gap that the laboratory technician must close.

Aviation fuel testing specifics. The fuel chemistry laboratory technician who targets the aviation fuel quality domain must understand that aviation fuel testing is governed by a specific combination of international standards (ASTM D1655, DEF STAN 91-091), Joint Inspection Group (JIG) quality management requirements, and aviation authority oversight that goes substantially beyond the quality framework of a generic petroleum testing laboratory. The JFTOT test, the WSIM test, and the specific cleanliness requirements for aviation fuel sampling equipment have no direct equivalent in other petroleum products testing.

By Career Stage

Early Career Laboratory Technician (0–4 years)

The recently qualified laboratory technician has good basic measurement and documentation skills and the most flexibility in terms of which aerospace domain to target. The preparation task is primarily regulatory and domain-specific: understand the NADCAP framework for the target domain, identify and pursue the relevant certification pathway (ASNT or ISO 9712 for NDT, NADCAP NLT for materials testing), and build at least one demonstrable aerospace domain engagement through self-study.

Experienced Laboratory Technician (5–15 years)

The experienced laboratory technician has developed the measurement confidence, equipment fluency, and quality system instincts that aerospace values most. The preparation task is specifically regulatory: mapping existing competencies to the aerospace qualification framework, identifying the specific additional certifications required, and demonstrating the ability to connect their measurement work to its engineering consequence. Mid-career technicians with specialist expertise in a specific test method (fatigue testing, SEM metallography, fuel analysis) are competitive for specialist roles at aerospace test laboratories from a relatively early stage of their transition preparation.

Senior Laboratory Technician / Laboratory Supervisor

The senior laboratory technician who has managed test programmes, written test procedures, and supervised junior staff has competencies that map directly to laboratory team leader and quality assurance roles in aerospace test facilities. The regulatory vocabulary gap is the primary obstacle — senior technicians who specifically address the NADCAP accreditation framework, the aerospace material specification system, and the quality consequence of aerospace test data are strongly competitive for experienced technician and team leader roles.

Domain Mapping — Where Your Laboratory Background Fits Best

Mechanical Testing Technician → Aerospace Materials Testing, Mechanical Properties Laboratory

Test machine operation, specimen preparation, load-extension data analysis, report writing — transfer directly with aerospace standards adaptation. The aerospace addition: AMS and ASTM method selection for specific aerospace materials, NADCAP NLT accreditation requirements, A-basis and B-basis statistical significance of test population size, and the specific specimen preparation tolerances required for aerospace allowables development.

Metallography / Microscopy Technician → Failure Investigation, Microstructure Qualification

Sectioning, mounting, grinding, polishing, etching, imaging — transfer directly. The aerospace addition: aerospace alloy microstructure acceptance criteria (grain size per ASTM E112 for aerospace wrought alloys, alpha-case depth limits for titanium, recrystallisation requirements for nickel superalloys), failure investigation reporting under the aerospace occurrence reporting framework, and ASTM E3 metallographic preparation standards for aerospace materials.

NDT-Experienced Technician → Aerospace NDT Level I/II, MRO Inspection, Composite Inspection

Physical method knowledge, probe handling, indication interpretation — transfer with formal certification. The aerospace addition: ASNT SNT-TC-1A or EN ISO 9712 formal certification, aerospace-specific acceptance standards (ASTM E2375 for UT of composites, AMS 2640 for ET of engine components), and the damage tolerance link between NDT sensitivity and inspection interval.

Chemical Analysis Technician → Aviation Fuel Quality, Propellant Analysis

Analytical technique execution, standard preparation, instrument calibration, report writing — transfer with aviation-specific method adaptation. The aerospace addition: ASTM D1655 and DEF STAN 91-091 for aviation fuel, JIG Quality Management requirements, JFTOT and WSIM test methods, and SAF qualification under ASTM D7566.

Calibration / Metrology Technician → Aerospace Calibration Management, CMM Operation

Calibration procedure execution, uncertainty estimation, calibration certificate production — transfer with aerospace traceability standard. The aerospace addition: AS9100 calibration management requirements, NADCAP NLT calibration laboratory scope, AS9102 First Article Inspection requirements, and GD&T (Geometric Dimensioning and Tolerancing) for aerospace part inspection.

Environmental Test Technician → DO-160 Test House, Avionics Environmental Qualification

Test chamber operation, data acquisition, test report writing, specimen handling — transfer with aerospace regulatory standard. The aerospace addition: RTCA DO-160 (the governing standard for environmental testing of airborne equipment) sections for temperature, altitude, vibration, humidity, shock, and EMI, and the test witness and report requirements for aviation authority acceptance.

Quality Control Laboratory Technician → Aerospace Receiving Inspection, First Article Inspection

Inspection procedure execution, measurement equipment use, non-conformance reporting — transfer with aerospace quality standard. The aerospace addition: AS9100 quality management, AS9102 First Article Inspection Report requirements, AMS material specification acceptance criteria, and the Part-21 approved design organisation interface for engineering dispositions.

PART II: 20 SPECIFIC PREP TASKS FOR THE TRANSITION

FOUNDATIONAL — DO THESE FIRST

1. Understand NADCAP — what it is, what it covers, and what it requires that ISO 17025 does not

4–5 hours

Skill: The aerospace laboratory accreditation framework — the most important quality credential for laboratory-based aerospace roles

Why the field cares: NADCAP is the National Aerospace and Defense Contractors Accreditation Program — the industry-managed accreditation programme that validates whether special process suppliers (including materials and NDT testing laboratories) meet the requirements of aerospace prime contractors and government customers. For laboratory technicians, the most relevant NADCAP commodity is NLT (Non-Destructive Testing and Laboratory Testing), which covers mechanical testing laboratories, chemical testing laboratories, metallography laboratories, and NDT laboratories performing work for aerospace customers. The key difference from ISO 17025: NADCAP NLT requires specific calibration standards (ASTM E4 for force verification of testing machines is required, not just any accredited calibration), specific reference standard traceability (hardness test blocks traceable to NIST or equivalent national standard), specific test report content requirements (every test report must identify the specification revision, the test method revision, the equipment used and its calibration status, and the acceptance criteria), and specific personnel qualification requirements that ISO 17025 does not.

4–5 hours

Done signal: You can describe at least four specific NADCAP NLT requirements that go beyond ISO 17025 — specifically: the ASTM E4 force verification requirement, the traceability requirement for reference standards, the test report content requirement, and the corrective action process requirement (that every NADCAP audit finding must be investigated to root cause with a corrective action plan submitted to the PRI within 30 days of the audit). You have identified whether any current employer holds NADCAP accreditation and what commodity it covers.

4–5 hours

2. Understand the AMS specification system — how to read an AMS specification and apply it to a test requirement

Skill: The material specification framework that governs every aerospace test requirement

Why the field cares: Aerospace materials testing is performed to specific AMS (SAE Aerospace Material Specifications) or ASTM specifications, and understanding which specification governs a specific test is the regulatory literacy that aerospace test laboratory technicians must have. When a job traveller (the work order that accompanies an aerospace specimen through the test laboratory) specifies “tensile test per AMS 4928 paragraph 3.5.1,” the technician must be able to locate AMS 4928, identify what the specification requires for tensile testing (specifically which ASTM method is called out — typically ASTM E8 for room temperature tensile — and what the specimen type and dimensional requirements are), and identify the acceptance criteria (the minimum yield strength, ultimate tensile strength, elongation, and reduction of area values that the test result must meet). The failure to identify the correct acceptance criteria — testing to the correct method but comparing results against the wrong specification revision or the wrong condition — produces a test report that is technically executed correctly but commercially and regulatorily invalid.

Done signal: You can navigate an AMS specification for a specific aerospace material — for example, AMS 4928 for Ti-6Al-4V bar — and identify: the scope and applicability, the composition limits, the heat treatment condition required, the specific mechanical property tests and the ASTM method called out for each, the specimen type and dimensional requirements, and the acceptance criteria for each property. You can explain the difference between the “shall” requirements (mandatory, tested for every lot) and the “should” requirements (recommended practice, not necessarily tested for every lot).

4–5 hours

3. Understand the five primary NDT methods — physical principles, aerospace applications, and formal certification pathways

Skill: The non-destructive testing foundation for the largest category of aerospace laboratory roles

Why the field cares: NDT — the inspection of materials and components for defects without damaging them — is the largest single category of laboratory-based aerospace employment. Dye penetrant (PT), magnetic particle (MT), ultrasonic (UT), eddy current (ET), and radiographic (RT) testing together employ thousands of technicians across aerospace MRO, OEM manufacturing, and specialist inspection services. The laboratory technician who has used any of these methods in a quality control context — even informally — has practical exposure that formal certification training can build on. The formal certification pathways are: ASNT SNT-TC-1A (employer-based certification, widely used in the US and internationally) or EN ISO 9712 (third-party certification, required for EASA-regulated NDT in Europe). Each method requires separate certification. The methods most accessible from a laboratory background without prior NDT exposure — because the physical equipment is simpler and the technique more directly transferable from general laboratory practice — are PT (which requires specimen preparation skills and visual judgement similar to metallographic examination) and UT (which requires the systematic test technique and data recording discipline of mechanical testing).

Done signal: You can describe the physical principle, the primary aerospace application, and the formal certification path for each of the five main NDT methods. You understand the difference between ASNT SNT-TC-1A and EN ISO 9712 — specifically that SNT-TC-1A is employer-administered while EN ISO 9712 requires independent third-party examination — and know which is required for

the roles you are targeting. You have identified a specific ASNT or ISO 9712 certification training programme for your primary target NDT method (PT or UT for most laboratory technicians without prior NDT) and know the training hour requirements and examination format.

4–5 hours

4. Understand ISO 17025 at the level of its key requirements and how they connect to aerospace quality

2–3 hours

Skill: The laboratory accreditation standard that most laboratory technicians already work under — and its aerospace significance

Why the field cares: ISO/IEC 17025:2017 (General requirements for the competence of testing and calibration laboratories) is the foundation of accredited laboratory practice. The laboratory technician who has worked in an ISO 17025-accredited laboratory already understands — at a practical level — scope of accreditation (what tests the laboratory is accredited to perform, and what tests require a caveat or are outside the scope), method validation (the process of demonstrating that a test method is fit for purpose), measurement uncertainty (the quantification of the combined effect of all sources of error in a measurement, reported as an expanded uncertainty at a stated confidence level), and the corrective action system (the formal process for investigating non-conforming test results and implementing corrective actions). Articulating this understanding in an aerospace interview — connecting the ISO 17025 requirements the technician already follows to their aerospace equivalent — is the quality management translation that makes the technician's laboratory experience legible to an aerospace quality manager.

Done signal: You can explain what scope of accreditation means and why it matters for an aerospace customer (the accreditation certificate specifies exactly which test methods the laboratory is accredited to perform; a test performed outside the scope, even by a competent technician, is not an accredited test and cannot be referenced as such in an aerospace test report). You can describe what measurement uncertainty means — not the full mathematical framework, but the practical concept: that every measurement has a range of values within which the true value is likely to lie, and that the width of that range must be known to determine whether a measured value that is close to the specification limit actually passes or fails. You can describe the corrective action process in your current laboratory — what triggers a corrective action, who is responsible for investigating, and what documentation is produced.

5. Understand aerospace quality management — AS9100, NADCAP, and the Part-21 Design Organisation interface

4–5 hours

Skill: The quality management framework that governs aerospace manufacturing and testing

Why the field cares: AS9100 is the aerospace extension of ISO 9001 that adds requirements specific to the aerospace sector: risk management for quality, configuration management of design documents and approved material specifications, key characteristic identification and control, first article inspection under AS9102, counterfeit part prevention, and the specific documentation requirements for aerospace product conformance. The laboratory technician who has worked under ISO 9001 has the foundational quality management understanding; the AS9100 additions need to be specifically studied. Additionally, for laboratory technicians working at OEM materials test laboratories or test laboratories connected to the design and certification process, understanding the Part-21 Design Organisation Approval (DOA) interface is important: when a material test result is used to support a structural design allowable or to close a design deviation, the test result becomes part of the DOA's compliance evidence and must meet specific documentation and traceability requirements.

Done signal: You can describe three AS9100 requirements that add to ISO 9001 in ways specifically relevant to laboratory testing — the key characteristic requirement, the first article inspection requirement under AS9102 (that the first production part of a new part number must have every drawing dimension verified and documented), and the counterfeit part prevention requirement (that material certification documents must be verified for authenticity, and suspect counterfeit material must be quarantined and reported). You can explain what a Part-21 DOA is and why a test result used

as compliance evidence for a type certificate must meet the DOA's specific documentation requirements.

4–5 hours

6. Learn the aerospace mechanical testing standards — ASTM E8 for tensile, E18 for Rockwell hardness, E23 for impact, and E606 for fatigue

4–5 hours

Skill: The specific ASTM test standards that govern aerospace mechanical properties testing

Why the field cares: The laboratory technician who has performed tensile, hardness, impact, and fatigue testing in a general commercial testing environment has the practical skills for aerospace materials testing — but they may have performed the tests to different ASTM revisions, different specimen geometries, or under different environmental conditions than the specific aerospace requirement. ASTM E8 (tensile testing of metallic materials) specifies allowable specimen geometries, grip alignment requirements, extensometer attachment requirements, and strain rate requirements that must all be met for the test to be valid for aerospace allowables derivation. ASTM E18 (Rockwell hardness) specifies indenter and anvil condition requirements, cycle time requirements, and reference block frequency of use requirements that affect result validity. ASTM E23 (notched bar impact testing — Charpy) specifies specimen preparation tolerances, striker geometry, and temperature conditioning requirements. ASTM E606/E606M (strain-controlled fatigue) specifies cycle waveform, control mode switching procedures, and failure criterion definition.

Done signal: For at least two of the four test methods (E8, E18, E23, E606), you can state: the specimen type options and their dimensional tolerances, the key procedural requirement that most commonly causes aerospace test failures when not correctly followed (for E8: extensometer gauge length and attachment without notching the specimen; for E18: reference block frequency of use and minimum hardness value distance from specimen edge; for E23: striker geometry verification and specimen temperature conditioning time), and the acceptance criterion for at least one aerospace alloy tested to that method.

7. Understand calibration management for testing equipment — the traceability chain, calibration intervals, and out-of-calibration consequences

3–4 hours

Skill: The quality management of measurement — and the aerospace-specific consequences of calibration failure

Why the field cares: The laboratory technician who has managed or worked within a calibrated equipment system understands calibration in principle. The aerospace-specific dimension is the consequence of out-of-calibration equipment: in a commercial testing laboratory, an out-of-calibration test machine means the test results produced since the last valid calibration are uncertain and may need to be repeated. In an aerospace materials testing context where those results have been used to establish design allowables incorporated into a structural analysis, an out-of-calibration discovery triggers a formal engineering investigation (was the calibration drift significant? did any reported results fall within the calibration error range of an acceptance limit?), a corrective action programme, and potentially a formal engineering departure or airworthiness concern. The NADCAP NLT auditor specifically checks that calibration records are complete, that equipment found out of calibration triggers an impact assessment on affected test results, and that the impact assessment is documented and closed with appropriate engineering authority.

Done signal: You can describe the complete out-of-calibration response process in an aerospace test laboratory — the identification of out-of-calibration equipment, the quarantine of the equipment, the identification of all test results produced since the last valid calibration, the assessment of whether any out-of-calibration condition could have affected a result that was close to an acceptance limit, the documentation of the assessment and its conclusion, the notification to the customer if affected test results were already reported, and the root cause investigation and corrective action. You can explain what “as-found condition” means on a calibration certificate and why it is specifically important for aerospace.

4–5 hours

8. Understand test specimen preparation for aerospace mechanical testing — tolerances, surface finish, and contamination control

Skill: The precision preparation that determines whether a mechanical test result is valid

Why the field cares: Specimen preparation is the step in the mechanical testing process that is most directly analogous to the laboratory technician's existing sample preparation skills — and the step where aerospace-specific requirements add the most specific constraints. An aerospace tensile specimen machined to ASTM E8 proportional specimen dimensions must have: a gauge length that is within $\pm 0.5\%$ of the specified dimension; a gauge diameter that is measured to ± 0.001 inch and used to calculate the actual cross-sectional area for the stress calculation; a surface finish in the gauge section that meets the Ra requirement (typically $\leq 32 \mu\text{in}$ for metallic tensile specimens) to avoid surface-initiated premature failure from machining marks; a fillet radius between the gauge section and the grips that is no less than the specified minimum to avoid stress concentration at the grip transition. A specimen that fails in the grip section — regardless of the fracture load — is an invalid test because the fracture location indicates that stress concentration rather than material properties governed the failure.

Done signal: You can describe the dimensional tolerance requirements for at least one aerospace tensile specimen type (for example, the ASTM E8 0.500-inch round specimen: gauge diameter 0.500 ± 0.010 inch, gauge length 2.000 ± 0.005 inch, fillet radius minimum $3/8$ inch), the consequence of each out-of-tolerance condition on the test result (gauge diameter error causes proportional error in calculated stress; surface finish violation causes fatigue-like premature failure in a static tensile test), and the acceptance criterion for fracture location (fracture within the gauge section is acceptable; fracture outside the gauge section invalidates the test regardless of the load value). You have experience machining or preparing test specimens and can describe the practical challenges of achieving the specified surface finish on hard aerospace alloys (titanium, Inconel).

2–3 hours

9. Prepare specific technical stories — measurement problem accounts with the Situation-Action-Result structure

Skill: The interview narrative that demonstrates measurement competence through specific experience

Why the field cares: Aerospace laboratory interviews are behavioural in structure — they probe real experience through specific scenario questions because the evidence of measurement competence is the ability to describe specific technical situations with the accuracy that genuine experience provides. The laboratory technician's version of the fault-finding story is the measurement anomaly story: "tell me about the most unusual or anomalous test result you have encountered and how you investigated it." The weak answer: "I had some results that were lower than expected and we retested." The strong answer: "We had a batch of EN8 tensile specimens that were consistently measuring yield strength about 15% below the specification minimum, which was unusual because the material certificate showed the composition was in specification. I checked the extensometer calibration — it was current. I rechecked the specimen dimensions — gauge section was within tolerance. I then measured the hardness on the untested specimens and found it was about 5 HRC below what EN8 in the heat-treated condition should be. I checked the heat treatment record and found the furnace temperature chart showed a 30°C deviation during the soak period. The specimens were tested from an improperly heat-treated batch. The batch was quarantined and the heat treatment supplier was notified." This level of investigative specificity — connecting a test anomaly through the measurement system back to a process failure — is what aerospace laboratory employers look for.

Done signal: You have three genuine measurement anomaly or investigation stories from your laboratory experience, each with the Situation (the material, the test, the anomalous result), the Action (the investigative steps taken, in sequence, with the specific measurements and observations at each step), and the Result (the root cause identified, the disposition of the affected material or specimens, and the corrective action). Each story demonstrates systematic investigation — tracing the anomaly through the measurement system to the physical cause — rather than unexplained variation accepted as normal scatter.

10. Build a demonstrable connection to an aerospace-specific test or inspection method — through self-study and practical access 20–40 hours

Skill: The portfolio engagement that differentiates genuine aerospace preparation from generic laboratory interest

Why the field cares: The laboratory technician's portfolio project is different from the engineering candidate's portfolio project — it is less about building something from scratch and more about demonstrating engagement with a specific aerospace test method or specification at a level of practical familiarity that general laboratory experience does not provide. The most accessible demonstrable connections: completing an ASNT Level I or II PT or UT training course; studying the DO-160 standard and describing at least one complete environmental test category (temperature, vibration, or humidity) at the level of its specific test conditions, acceptance criteria, and the physical equipment required; or reviewing the JIG (Joint Inspection Group) manual for aviation fuel quality and describing the specific analytical tests required for a Jet A-1 quality check at a major airport. Each of these is achievable in 20–40 hours of focused self-study and produces specific technical content for the interview that distinguishes a prepared candidate from an interested one.

Done signal: You have engaged at practical depth with at least one specific aerospace laboratory or inspection method: either completed the formal training course for ASNT Level I/II certification in one method (PT or UT), or studied the DO-160 standard sufficiently to describe a complete temperature test (Category A2) including the test conditions, the specimen soak duration, the performance check requirements, and the acceptance criteria, or studied the JIG Guidelines enough to describe the complete quality check sequence for a refuelling from an airport hydrant system including the filter monitor inspection, the bottom sample procedure, and the density and appearance checks. You can discuss your chosen method at this level of specificity for 10 minutes without notes.

ROLE-SPECIFIC — TARGET AFTER FOUNDATIONAL TASKS**11. For NDT roles: understand what aerospace NDT Level II certification requires and commit to the training pathway** 10–15 hours

Skill: The specific certification pathway for the largest aerospace laboratory role category

Why the field cares: ASNT SNT-TC-1A Level II certification in the UK and international markets, or EN ISO 9712 Level II certification for EASA-regulated work, is the required professional credential for independent NDT work in aerospace. The certification requirements are specific: Level II requires a minimum number of training hours (ASNT: PT 8 hours general study + 8 hours specific, UT 40 hours general + 40 specific; ISO 9712 has similar minimums), a minimum number of practical experience hours (ASNT: PT 400 hours, UT 800 hours, RT 1,500 hours before Level II examination eligibility), written examinations (general, specific, and practical components), and annual certification by the employer's Level III. The laboratory technician who has used PT or UT in a quality control context has practical exposure but may not have the documented training hours that count toward certification. The most effective route: enroll in an approved Level II training course (typically a 2-week intensive programme for PT, a 3–4 week programme for UT), which provides the required training hours, prepares for the written examinations, and provides a practical test on reference standards.

Done signal: You have identified a specific ASNT Level II or EN ISO 9712 Level II training course for your target primary method (PT or UT for most laboratory technicians), know its cost and duration, have verified that the training provider is approved by ASNT or an ISO 9712-accredited certification body, and have a plan for accumulating the required practical experience hours after training (either in your current role if PT or UT is used there, or in a transitional role at an MRO or inspection services company). You have studied the full NDT Technician guide in this series, which provides detailed method descriptions and interview preparation for each method.

12. For materials testing roles: understand the NADCAP NLT audit scope and common audit findings 3–4 hours

Skill: The specific quality management challenges of a NADCAP-accredited aerospace test laboratory

Why the field cares: The NADCAP NLT audit — conducted by a Performance Review Institute auditor at the laboratory — reviews every aspect of the laboratory's quality management system against the NADCAP AC7101 checklist (for mechanical testing) or AC7114 (for NDT). The most frequently cited non-conformances in NADCAP NLT audits — the findings that most commonly cause aerospace test laboratories to receive a conditional rather than full audit result — are: calibration records not documenting the as-found condition of equipment before adjustment (so the impact of the out-of-calibration period cannot be assessed); test reports not citing the specification revision (so the customer cannot confirm the correct revision was applied); specimen preparation records not documenting the actual measured dimensions used in the calculation (so the cross-sectional area used cannot be independently verified); and qualification records for testing personnel not demonstrating that they have been specifically trained and evaluated on the specific test methods they perform.

Done signal: For each of the four common NADCAP NLT audit findings above, you can explain: what the finding represents (what quality failure occurred or could have occurred because the requirement was not met), what the correct practice is, and what the consequence for the customer would be if the finding represented an actual quality failure rather than merely a documentation gap. You can describe the NADCAP corrective action process — the requirement to submit a root cause analysis and corrective action plan within 30 days of the audit close, and the risk of suspension of NADCAP accreditation if the corrective actions are not implemented within the agreed timescale.

13. For fuel quality roles: understand the aviation fuel testing sequence and JIG quality management 5–6 hours

Skill: The specific analytical chemistry and quality management of aviation fuel

Why the field cares: Aviation fuel quality is a specialised application of petroleum product analytical chemistry that is governed by a specific combination of ASTM standards (primarily ASTM D1655 for Jet A and Jet A-1), DEF STAN 91-091 for UK and NATO applications, and the JIG (Joint Inspection Group) Guidelines for fuel handling quality management at airports. The laboratory technician who has performed general petroleum products testing — flash point, density, viscosity, distillation — has the analytical technique foundation. The aviation-specific additions are: the JFTOT (ASTM D3241) thermal oxidative stability test and its relationship to fuel nozzle coking in aircraft engines; the WSIM (ASTM D3948 or D2550) water separation index test and its relationship to water contamination management in aircraft fuel systems; the microbiological contamination test (ASTM D6469) for detecting bacterial growth in aircraft fuel storage; and the specific sampling protocols and chain of custody documentation that JIG requires for aviation fuel quality control.

Done signal: You can describe the JFTOT test procedure at the level of what it measures (the tendency of the fuel to form deposits on a heated tube at 260°C, which simulates the nozzle environment in an aircraft fuel injection system), what result it produces (the tube deposit rating on a visual scale of 1–4, with 1 being clean and 4 being heavy deposit, and the pressure differential across the filter pad), and what the acceptance criterion for Jet A-1 is (tube deposit rating ≤ 1 and filter pressure differential ≤ 25 mm Hg for ASTM D1655 Annex A1 quality requirements). You can describe the JIG chain of custody requirement for aviation fuel sampling — the specific sampling protocol, the labelling requirements, and the documentation that must accompany a fuel sample from the point of origin through the testing laboratory to the test report.

14. For environmental test roles: understand DO-160 — the environmental test standard for airborne equipment 5–6 hours

Skill: The governing test standard for environmental qualification of avionics — the regulatory basis for every environmental test in aerospace avionics certification

5–6 hours

Why the field cares: RTCA DO-160 (Environmental Conditions and Test Procedures for Airborne Equipment) is the standard that defines the environmental tests that avionics equipment must pass before it can be certified for installation in aircraft. A DO-160 test facility — the specialist environmental test laboratories that certification bodies recognise — tests avionics LRUs (Line Replaceable Units) against the DO-160 test categories: temperature (operating low and high, short-term operating, temperature variation), altitude (operating and storage), humidity, vibration (sinusoidal and random), shock, acoustic noise, EMI (conducted susceptibility and emission, radiated susceptibility and emission), power input (voltage and frequency variations, transients), and others. The laboratory technician at a DO-160 test facility sets up the test specimen, connects the monitoring instrumentation, runs the test programme per the approved test procedure, records the performance throughout, and produces the test report that the certification authority accepts as compliance evidence.

Done signal: You can describe the DO-160 temperature test — what Section 4 (Temperature and Altitude) requires (the specific test temperatures for the category assigned to the equipment, the soak duration, the performance monitoring required during the test, and the criteria for confirming the equipment has reached thermal equilibrium before performance testing begins). You can explain what the DO-160 random vibration test (Section 8) simulates (the broadband vibration environment of the aircraft installation location — fuselage, engine-mounted, or rotary-wing), what the power spectral density (PSD) profile means physically (the vibration energy per unit frequency as a function of frequency), and why the test duration is set to simulate the accumulated vibration exposure over the equipment's design life. You have identified a DO-160 test facility in your target area and researched their specific test capabilities and typical customer base.

4–5 hours

15. For composite process roles: understand the NADCAP composite fabrication audit requirements and autoclave process control

Skill: The process quality management of aerospace composite manufacturing — the most demanding composite quality environment

Why the field cares: The NADCAP composite fabrication accreditation (NADCAP AC7118) covers the manufacturing processes by which composite parts are made: prepreg lay-up, autoclave cure, vacuum infusion, and wet lay-up. For the laboratory technician, the most accessible entry point is not the manufacturing floor but the process control laboratory: the environment monitoring (temperature, humidity, and cleanliness of the lay-up area), the prepreg incoming inspection (gel time, resin content, and tack testing), the cure cycle monitoring and documentation (thermocouple placement and data recording during cure), and the finished part inspection (ultrasonic C-scan for delamination, void content by optical metallography of test panels, and mechanical property testing of witness panels). Understanding what AC7118 requires for each of these activities — the specific calibration requirements for cure cycle thermocouples, the specific incoming inspection tests for prepreg, and the specific documentation requirements for first article testing of composite parts — is the composite process quality management knowledge that NADCAP-accredited composite manufacturers need from their quality and process control laboratory technicians.

Done signal: You can describe the NADCAP AC7118 requirement for autoclave cure cycle monitoring — the requirement for thermocouple calibration (all thermocouples must be calibrated at least annually using a NIST-traceable standard), the number of thermocouple required per cure run (at least two, positioned to monitor the coldest and hottest locations on the part), the survey requirement (the autoclave temperature distribution must be characterised by a loaded survey with the maximum number of thermocouples the autoclave can accommodate, to identify hot and cold zones), and the record requirement (the time-temperature record for every cure cycle must be retained as a quality record). You can describe what a prepreg incoming inspection test involves — the specific tests typically performed (gel time, tack, and resin content), what each test measures, and what the out-of-specification consequence is (the prepreg lot is quarantined and the customer is notified, because out-of-specification prepreg may not cure to the required properties).

5–6 hours

16. Understand aerospace first article inspection — AS9102 requirements and the CMM measurement process**Skill:** The dimensional inspection process that verifies the first production part of a new design**Why the field cares:** AS9102 (Aerospace First Article Inspection Requirement) is the standard that requires the first production part of every new aerospace part number to be completely inspected — every dimension, every material property, and every functional characteristic specified on the drawing must be measured and recorded in the First Article Inspection Report (FAIR). The FAIR is a permanent quality record that demonstrates the manufacturing process can produce a part that meets the design specification. For the quality and metrology laboratory technician, the FAIR process involves: reading and interpreting aerospace engineering drawings with GD&T (Geometric Dimensioning and Tolerancing) callouts, measuring every dimension using appropriate measurement equipment (CMM for complex three-dimensional features, height gauge and surface plate for simpler dimensions, go/no-go gauges for thread and bore features), recording the measured value against the drawing tolerance and noting any deviation, and assembling the complete FAIR package with the drawings, the measurement data, the material test report, and the special process certifications.**Done signal:** You can describe the three main sections of an AS9102 First Article Inspection Report (the design characteristic accountability — listing every drawing requirement; the actual measurement data — the measured value for each characteristic; and the supporting documentation — material certificates, process certifications, and test reports). You can explain what a GD&T position tolerance of $\text{⊗ } \text{Ø}0.010 \text{ A B}$ means (the centre of the feature must fall within a cylindrical zone of diameter 0.010 inch located by the basic dimensions from datum references A and B). You can describe the difference between a CMM (Coordinate Measuring Machine) measurement and a conventional measurement — specifically that a CMM can measure the full geometric form of a feature from a point cloud, while a conventional measurement captures only one or a few dimensions.

3–4 hours

17. Understand the aerospace occurrence reporting and non-conformance management system**Skill:** The quality reporting framework that governs every non-conformance in aerospace manufacturing and testing**Why the field cares:** Every deviation from specification in aerospace manufacturing or testing — a dimension out of tolerance, a material property below the acceptance minimum, a process step performed incorrectly — must be formally documented, investigated, and dispositioned through the organisation's non-conformance management system. The non-conforming item is tagged with a Rejection Tag or Non-Conformance Report (NCR), quarantined from use in production or further processing, and held pending a disposition decision. The disposition options are: use-as-is, rework, repair, or scrap. For the laboratory technician, non-conformances arise most commonly from: test results that fall outside the acceptance criteria, calibration failures that invalidate previous test results, and specimen preparation errors that require the specimen to be remade. Understanding how to correctly document a non-conformance — specifically what information the NCR must contain (the part number and serial or lot number, the specification requirement, the actual finding, the location or test step where the non-conformance was found, and the immediate containment action) — is the quality management knowledge that every aerospace laboratory technician must demonstrate.**Done signal:** You can describe the NCR process in aerospace quality management: what information the NCR must contain, what the three formal disposition options are (use-as-is, rework/repair, and scrap), what level of engineering authority is required to approve a use-as-is disposition, and what happens to parts that are found non-conforming during a NADCAP audit. You can describe a specific non-conformance you have documented in your laboratory experience — what was found, what documentation you produced, and what the disposition was.

3–4 hours

18. Build quantitative estimation fluency for laboratory measurements — approximate ranges and order-of-magnitude checks without reference tables**Skill:** The measurement awareness that experienced aerospace laboratory technicians develop

3–4 hours

Why the field cares: The experienced aerospace materials testing or NDT technician is expected to have enough working knowledge of typical material property ranges to recognise immediately when a result is anomalous — before the supervisor has reviewed the test report. A hardness reading of 25 HRC on a Ti-6Al-4V part specified to 28–36 HRC is worth flagging before the next test. A tensile strength of 850 MPa on a 2024-T3 aluminium specimen specified for minimum 420 MPa UTS should prompt an immediate check of whether the specimen or test machine is correctly identified. A penetrant indication with linear dimensions of 8mm in a weld with a 5mm maximum linear indication acceptance criterion requires immediate attention before the inspection is signed off. These quick sanity checks — the ability to know whether a result is in the right ballpark before detailed analysis is needed — are the measurement awareness that every experienced aerospace laboratory employer looks for in their technicians. They cannot be replaced by looking up tables; they come from familiarity with the materials and specifications being tested.

Done signal: You can state the approximate typical property ranges (to be verified against the specific specification before use) for at least five common aerospace materials tested in your target laboratory role — for example: Ti-6Al-4V bar annealed: approximately 900–1000 MPa UTS, 830–930 MPa 0.2% proof, 10–14% elongation; 7075-T6 aluminium sheet: approximately 500–570 MPa UTS, 430–500 MPa 0.2% proof, 8–11% elongation; IN718 bar solution annealed and aged: approximately 1380–1450 MPa UTS, 1170–1240 MPa 0.2% proof, 12% elongation minimum. You state these as approximate guidance values to be confirmed against the current approved specification before any acceptance decision.

ALL LEVELS

1–2 hours

19. Prepare a specific answer to "Why aerospace — and what specifically from your laboratory background makes you ready for this role?"

Skill: The transition motivation that converts the recruiter's uncertainty into recognition

Why the field cares: The aerospace laboratory recruiter who sees a laboratory technician CV — particularly one from a commercial testing, pharmaceutical, or food safety laboratory — has a specific initial uncertainty: is this person's measurement and documentation competence genuinely at the level aerospace requires, or has their previous environment been permissive enough that habits would need to be rebuilt? The answer must immediately and specifically demonstrate that the candidate understands what aerospace laboratory quality requires — not just that it is more demanding, but why. The answer "I have spent six years running tensile tests and I take pride in my documentation" is not sufficient. The answer "I have spent six years running tensile and hardness testing to ASTM E8 and E18, and the most significant thing I have learned is that a test result is only as reliable as the full chain of conditions that produced it — the specimen preparation, the calibration status of the machine and extensometer, the alignment of the specimen, and the stability of the test environment during the test. When I read about the NADCAP NLT requirement to document the as-found calibration condition before adjustment, I understood immediately why that matters — it is the only way to assess whether a result produced close to an acceptance limit was genuinely within specification or within the uncertainty range of an out-of-calibration machine. That connection between measurement quality and the engineering decision that depends on it is what I want to work on every day" is demonstrating the measurement philosophy that aerospace laboratory quality requires.

Done signal: You have a 90-second answer that: names the specific aerospace laboratory role and domain you are targeting (NDT, materials testing, fuel quality, environmental testing, or metrology); connects one specific aspect of your existing laboratory competence to an aerospace quality requirement at a level of technical specificity (naming a specific ASTM method, a NADCAP requirement, or a DO-160 test category); acknowledges the specific additional qualification or study you have done or are pursuing (ASNT certification, NADCAP NLT study, DO-160 standard review); and explains why the consequence of the work — specifically the fact that the test result informs an engineering decision with a safety consequence — is what motivates the aerospace application.

4–6 hours

20. Map your laboratory specialisation to specific aerospace employers and roles — and build a targeted application strategy**Skill:** The job market intelligence that focuses the transition on the highest-probability opportunities

Why the field cares: The aerospace laboratory job market is concentrated in specific geographic clusters — around major OEM and MRO facilities, around NADCAP-accredited test laboratory clusters (Sheffield and Bristol in the UK; Cincinnati, Hartford, and Seattle in the US; Toulouse and Hamburg in Europe), and around specialist test houses (DO-160 facilities are fewer in number but globally distributed in aerospace clusters). Understanding where the employers are, what their specific technical capability is, and what their hiring process looks like is the targeting exercise that makes the transition job search efficient. The laboratory technician who applies to “aerospace companies” without knowing whether the target employer has a materials test laboratory, what aerospace alloys they test, or what NADCAP commodity accreditation they hold is making applications that cannot be effectively prepared. The technician who applies to Element Materials Technology Sheffield for an aerospace tensile testing role, knowing that Element holds NADCAP NLT accreditation for mechanical testing, tests titanium and nickel alloys for Rolls-Royce and BAE Systems, and is actively recruiting to replace technicians who have progressed to supervisory roles, is making a targeted application that can be specifically prepared.

Done signal: You have identified at least five specific employers in your target aerospace laboratory domain and geographic area — including at least one NADCAP-accredited test laboratory, at least one MRO NDT inspection department, and at least one specialist test house or OEM laboratory. You have researched the specific capability of each, verified that your laboratory background is relevant to their work, and prepared applications for at least two — with a CV that explicitly connects your existing qualifications and experience to the NADCAP or DO-160 framework of the target employer, and a cover letter that names the specific test methods and materials you have experience with.

SUMMARY REFERENCE TABLE

#	Task Description
1 All / Do first	NADCAP — what it covers, what it adds to ISO 17025, PRI audit criteria
2 All	AMS specification system — reading a specification, method versus acceptance criterion
3 NDT target / All	Five NDT methods — principles, aerospace applications, certification pathways
4 All	ISO 17025 — key requirements articulated for aerospace relevance
5 All	AS9100, NADCAP, Part-21 DOA interface — aerospace quality framework
6 Materials testing	ASTM E8, E18, E23, E606 — specific requirements and failure modes for two methods
7 All	Calibration management — as-found condition, out-of-calibration consequences
8 Materials testing	Aerospace specimen preparation — dimensional tolerances, surface finish, fracture location
9 All	Measurement anomaly stories — three SAR narratives from existing laboratory experience
10 All / Critical	Demonstrable aerospace method engagement — ASNT training, DO-160 study, JIG review
11 NDT roles	ASNT / ISO 9712 Level II certification pathway — commit and plan for NDT target
12 Materials testing	NADCAP NLT audit scope and common findings
13 Fuel quality roles	Aviation fuel testing — JFTOT, WSIM, JIG quality management
14 Environmental test	DO-160 — temperature and vibration test sections at operational depth
15 Composite process	NADCAP AC7118 composite fabrication — cure cycle monitoring, prepreg incoming inspection
16 Metrology / quality	AS9102 first article inspection — FAIR structure, GD&T, CMM measurement
17 All	Aerospace NCR process — documentation, disposition options, authority levels
18	Approximate property ranges for common aerospace materials — order-of-magnitude awareness

#	Task Description
Materials testing	
19	“Why aerospace — what from laboratory background?” — 90-second specific answer
All	
20	Employer mapping — NADCAP database, EASA Part-145 scope, DO-160 test houses
All	

THE STRUCTURAL REALITY OF THE LABORATORY TECHNICIAN-TO-AEROSPACE TRANSITION

The laboratory technician entering aerospace is not moving between worlds — they are moving between quality regimes. The skills are the same. The instruments are recognisable. The physical techniques — specimen preparation, probe manipulation, instrument operation, data recording — are directly applicable. What changes is the consequence of getting it wrong, and the quality framework that exists specifically to prevent it.

In a commercial materials testing laboratory, an incorrect test result produces a rejected report, a retested specimen, and a corrected certificate. The customer is inconvenienced. In an aerospace materials testing laboratory, an incorrect tensile strength value for a lot of titanium bar that has already been machined into fan blade blanks and shipped to an engine manufacturer produces an engineering investigation, a potential fleet-wide airworthiness action, and potentially the discovery that the material used in a component currently installed on operating aircraft does not have the properties the structural analysis assumed. The test result that left the laboratory six months ago is now part of a chain of engineering decisions whose safety consequence is traceable back to the moment the technician chose the wrong acceptance criterion, failed to notice that the specimen gauge length was outside tolerance, or used an extensometer that had not been calibrated since its last inspection revealed an out-of-tolerance condition.

NADCAP exists because this chain of consequence is real and documented in aerospace accident investigations. The requirement to record the as-found calibration condition exists because a specific accident — or a specific fleet airworthiness concern — demonstrated that an out-of-calibration measurement produced results that were accepted as valid and used in a safety-critical engineering decision. The requirement to verify specimen dimensions and record the actual measured values exists because a specific test programme produced incorrect allowables from incorrectly prepared specimens, and those allowables were used to design a structure that subsequently failed in service.

The laboratory technician who understands this — who does not experience the NADCAP quality requirements as bureaucratic overhead but as the documented lessons of decades of aerospace quality failures applied to prevent the next one — is the technician who does the job that aerospace needs done. Not the one who produces the correct result by luck or by generally good practice, but the one who produces the correct result by systematic, documented, traceable, quality-managed process. Because the next test they run may be the one whose result ends up in an engineer's structural analysis, and the quality of that result — the completeness of the documentation, the currency of the calibration, the accuracy of the specimen preparation, the correctness of the acceptance criterion — matters in a way that no other laboratory context makes as immediately clear. That accountability, like the CRS signature of the licensed aircraft maintenance engineer, is what makes aerospace laboratory work specifically different from every other laboratory environment — and specifically worth the preparation this guide requires.

BREAK INTO AEROSPACE
